

A COMPUTATIONAL INVESTIGATION OF THE COLLATZ CONJECTURE

Affiliation: Euro International School, Sec-109

Address: Gurugram, Haryana

Email: kp3992929@gmail.com

1. Abstract

The Collatz Conjecture is a well-known mathematical problem which, although easy to describe, has still not been solved after many years. For this project I looked into the behaviour of Collatz sequences by means of a Python program. The program examined all the positive integers from 1 to 10,000 and noted how many steps it took to get down to 1 as well as the highest value attained in each sequence. The results indicated that the behaviour of a sequence is not entirely determined by its starting value. For instance, **6171** needed the most steps, namely **261**, while **9663** reached the highest value of **27,114,424**. The analysis also revealed a weak positive correlation of **0.2054** between the starting number and the stopping time. Even though this experiment does not prove the Collatz Conjecture, it does offer a helpful computational method for investigating its behaviour and for spotting interesting patterns.

2. Introduction

I am working on the Collatz Conjecture because it is a mathematical problem; even though the rules are simple and easy to understand the overall behaviour of the system is still not completely known. The process uses two steps: if the number is even it is divided by 2 and if the number is odd it is multiplied by 3 and then added by 1. Still with these rules mathematicians have not been able to show that every positive integer eventually reaches 1.

The problem is often connected to the mathematician Lothar Collatz. It is also called the $3n + 1$ problem. In this project I have not tried to prove or reject the conjecture. Instead, I wanted to use a computer to look at how different starting numbers act and to see if any clear patterns appear when many sequences are looked at together.

3. Background and Mathematical Foundation

Mathematically, the process can be written as:

$$T(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ 3n + 1, & \text{if } n \text{ is odd} \end{cases}$$

Each application of a rule is counted as one iteration. In this investigation, the **total stopping time** is defined as the number of iterations required for a starting value to reach 1:

$$S(n) = \text{number of iterations required for } n \text{ to reach } 1$$

The maximum value reached during a sequence is defined as:

$$M(n) = \max\{x_0, x_1, x_2, \dots, x_{S(n)}\}$$

where :

$$R(n) = \frac{M(n)}{n}$$

I also calculated the maximum-to-starting-value ratio:

This ratio will help us determine those sequences which grow farther away compared to their initial value. Each execution of the rule counts for one step or iteration. In this analysis, I was mostly concerned with two measures. First, there is the stopping time which refers to the number of iterations needed for the sequence to arrive at 1. Another measure which I used was $M(n)$ the maximum value attained by the Collatz sequence starting from the initial value until it gets to 1.

4. The Collatz Conjecture

The Collatz Conjecture starts with any positive integer n . If n is even, the next value is $\frac{n}{2}$. If n is odd, the next value is $3n+1$. These rules are then applied repeatedly to each new value.

The conjecture states that every positive integer will eventually reach 1 when this process is continued. After reaching 1, the usual rules produce the repeating cycle:

$$1 \rightarrow 4 \rightarrow 2 \rightarrow 1$$

For example, starting with 6 produces:

$$6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$$

This sequence needs eight steps to reach one. This sequence also climbs to a peak of sixteen. I find it interesting that single examples are easy to compute. Yet it is still a math problem to prove that this sequence always ends at one for every positive integer.

5. Research Question and Objectives

The main question I wanted to investigate was:

What patterns can be seen in the stopping times and maximum values of Collatz sequences for starting integers from 1 to 10,000?

To answer this I will record the stopping times for each starting integer. Note the maximum values reached during each Collatz sequence. The second part of my research was to record the ratio between the two to detect any unusual sequences where numbers went extremely far from their initial values.

The other goal was to check if bigger starting values would need more steps to reach 1. I also wanted to detect any unusual starting values which had significantly different behaviour from others. My goal was not to show that the conjecture was correct but to look at the conjecture carefully using information from a computer. I wanted to study the conjecture and see what the computer data said.

6. Methodology

I developed a Python program which tried out all the integers from 1 to 10,000. For each value, the program would apply the appropriate rule of Collatz until the number would get to 1.

As a part of each sequence, the program would calculate the number of steps and keep track of the highest number reached within that sequence. The data would be saved into a CSV file which would contain the number that was started with, stopping time and the highest number.

Using Python enabled me to calculate the same function for all 10,000 values at once without having to calculate each sequence individually.

7. Python Implementation

I created a Python program that follows the Collatz conjecture exactly. For each starting number I use three variables to track the value the number of steps and the maximum value reached.

The while loop runs until the current value becomes one. Inside the loop an if statement checks whether the number is even or odd. If the number is even I divide it by two. If the number is odd I apply the Collatz conjecture rule. Then I update the step counter and the maximum value.

I ran this process for every integer from one to ten thousand. I saved all the results, in a CSV data file. The entire code used in the analysis is provided in the Appendix.

8. Computational Experiments

The major experiment included an analysis of all starting values from 1 to 10,000. It generated a database with 10,000 different Collatz sequences. For each starting number, I was able to find out the stopping time and the maximum value of this sequence.

Afterwards, the data were sorted to determine the starting values that resulted in the longest sequences, the highest maximum values, and maximum to starting value ratios. Additionally, I was able to calculate the mean and median of all major parameters, including the correlation between starting value and stopping time.

Thus, I could analyze not particular examples but the behavior of all 10,000 sequences.

9. Results and Data Analysis

The results showed considerable variation between different starting numbers. The starting number **6171** had the greatest stopping time in the tested range, requiring **261 steps** to reach 1. The next highest was **9257**, which required **259 steps**, followed by **6943**, which required **256 steps**.

However, the sequence with the longest stopping time was not the sequence that reached the highest value. The starting number **9663** reached a maximum value of **27,114,424** and required **184 steps** to reach 1. This maximum was approximately **2,806** times its starting value.

The average stopping time for the 10,000 starting values was **84.97 steps**, while the median was **73 steps**. The mean maximum value was **58,996.48**, compared with a median of **13,336**. The difference between the mean and median is particularly noticeable for the maximum value, suggesting that a relatively small number of sequences reached exceptionally large values and increased the average.

The mean maximum-to-starting-value ratio was **11.79**, while the median was **3.38**. Again, the difference suggests that a relatively small number of unusual sequences had a large influence on the average.

10. Graphical Analysis

Graphs provide a useful way of visualizing the behaviour of all 10,000 sequences. A scatter plot of starting number against **total stopping time** shows whether the number of steps tends to increase as the starting number increases.

The data showed a weak upward tendency, but there was considerable variation between individual starting values. Some relatively small numbers required many more steps than nearby or larger numbers.

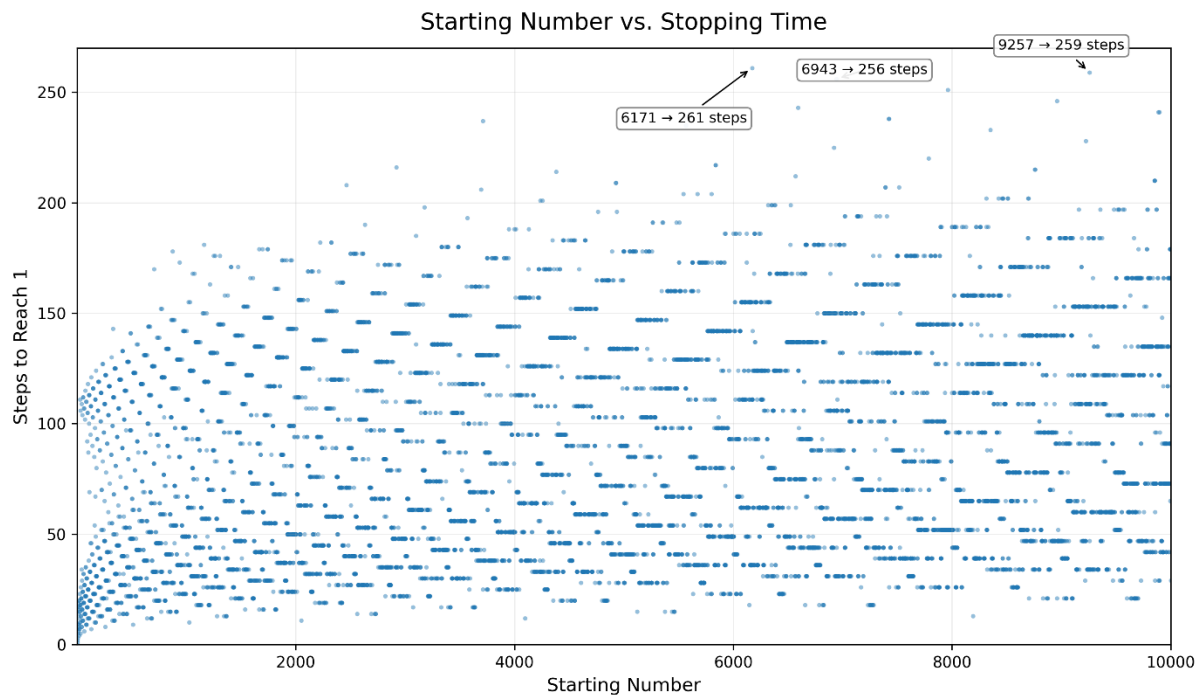


Figure 1: Starting number versus stopping time for all integers from 1 to 10,000.

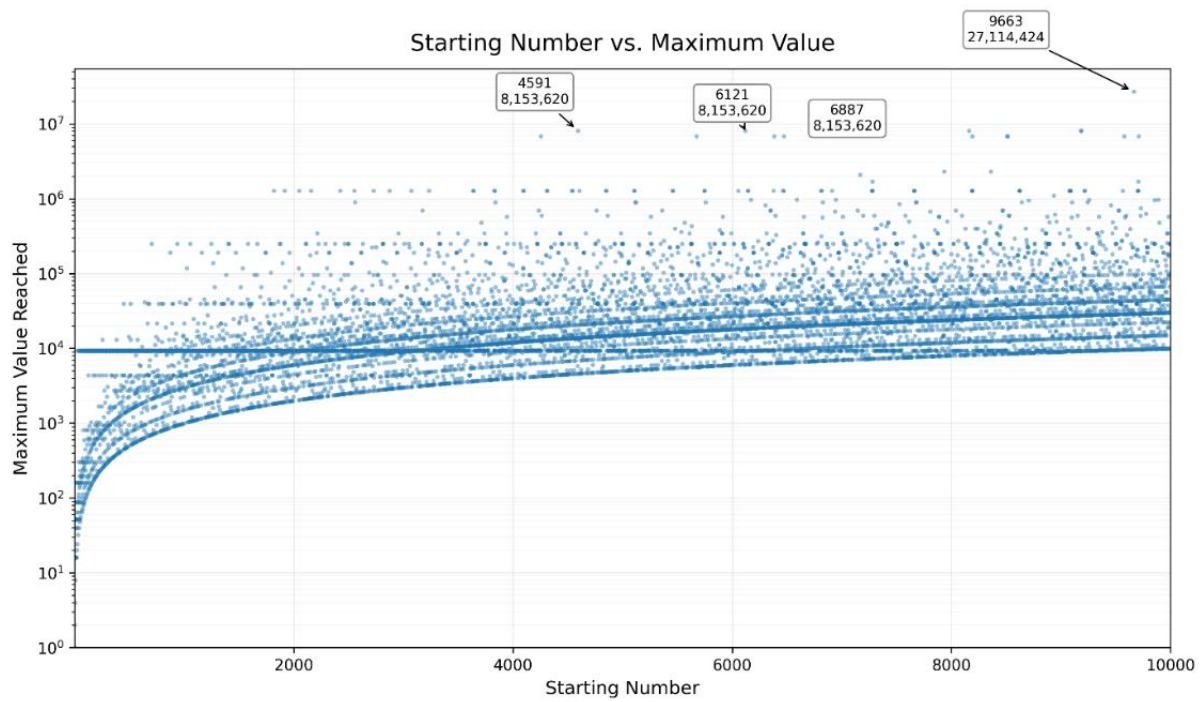


Figure 2: Starting number versus maximum value reached for all integers from 1 to 10,000.

Scatter plot between starting number and maximum value shows a lot of variations in terms of the growth of Collatz sequence. Logarithmic scale is used in Y axis because maximum values vary widely.

Result of **9663** is an outlier because of the fact that it grows to **27,114,424** although it starts from **9,663** only.

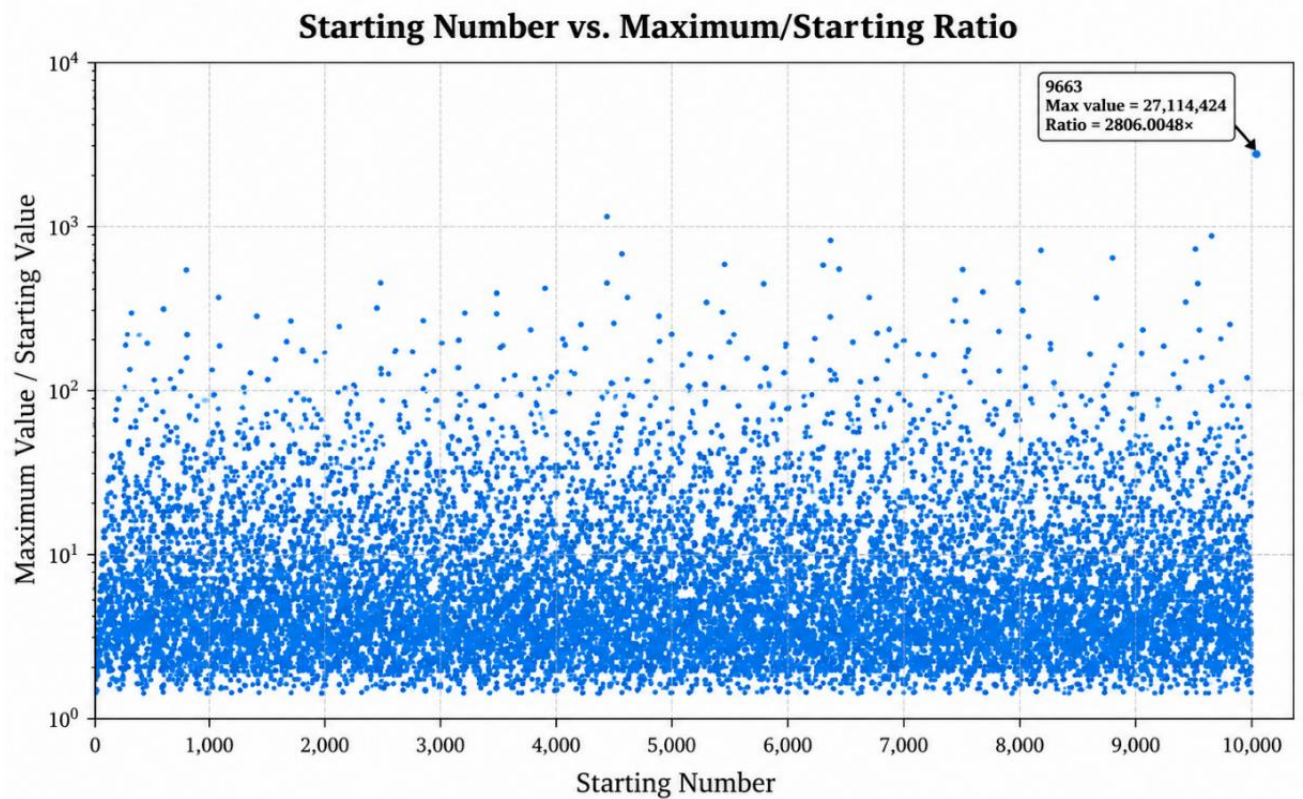


Figure 3: Starting number versus maximum-to-starting-value ratio for all integers from 1 to 10,000.

A third graph shows the **maximum-to-starting-value ratio**, highlighting sequences that grow especially far relative to their starting values. A logarithmic Y-axis is appropriate because the ratios vary substantially.

11. Patterns and Observations

One of the most obvious findings is that just because the initial value is large does not mean that it will have a large stopping time. For instance, the longest stopping time occurred in the case of **6171**, where it took **261** steps to reach 1, which means that it was the longest sequence in our range of tested numbers; however, there were many cases where a starting number was larger than **6171** and it took fewer steps to reach 1.

Also, the maximums were quite diverse among themselves. Thus, for instance, **9663** had a maximum of **27,114,424**, and there were several other numbers in the top 10 maximums that reached **8,153,620**.

Another curious finding is that some different initial values had the same maximum. In our experiments, there were several sequences with long stopping times that had the same maximum value of **975,400**. This occurs because different Collatz sequences can end up merging into one another at some point in the sequence. Once this happens, all transformations after this point are the same.

I can see that the particular sequence of odd transformations can influence the behaviour of the sequence.

12. Statistical Relationship

To investigate whether larger starting numbers generally required more steps, I calculated the Pearson correlation coefficient between starting number and total stopping time.

The Pearson correlation coefficient is given by:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

where x_i represents the starting number and y_i represents the corresponding total stopping time.

For the dataset containing all starting integers from 1 to 10,000, the resulting correlation was:

$$r = 0.205405$$

This represents a **weak positive linear correlation**. In other words, larger starting values tend to be associated with slightly larger stopping times on average, but the relationship is not strong.

The squared correlation is:

$$r^2 = (0.205405)^2 \approx 0.0422$$

This indicates that only about **4.2% of the variation in total stopping time is associated with the linear relationship with starting number**.

However, correlation does not imply causation and does not describe possible non-linear relationships. Therefore, starting number alone is not a reliable predictor of total stopping time.

13. Discussion

Moreover, the weak correlation of 0.2054 also reinforces the idea that the starting number itself cannot be considered a good indicator of the way a Collatz sequence will proceed. There is only a weak positive linear relationship between the starting number and the total stopping time. Even though higher values of starting numbers can be linked with longer sequences on average, the variations between the individual sequences are significant. Two starting numbers that differ from one another significantly may have sequences with very different total stopping times.

The value of the coefficient of determination $r^2 \approx 0.0422$ suggests that the linear model involving the starting number alone explains less than 4.2% of the variation in stopping time. It implies that the majority of the variation cannot be explained by the value of the starting number alone. Some other elements of the sequence are responsible for the observed pattern.

It is evident from the obtained results that the specific order of even and odd transformations is a vital element affecting the behaviour of the Collatz sequence. It explains why some relatively small starting numbers lead to unusually long sequences or extreme values of maximums.

14. Limitations

However, the major drawback of this study is the limited scope of starting numbers. The study was conducted using only numbers between 1 and 10,000, whereas the Collatz conjecture deals with an infinite number of positive integers.

Hence, the conclusion drawn from the fact that all starting numbers ended up being one can hardly be considered the proof of such conjecture. Computational research can be helpful in proving certain theories by providing additional evidence, but it can never substitute mathematical proof for the whole range of possible cases.

Finally, another drawback is the limited scope of this research: the study was conducted with the focus on stopping time and maximum value only, whereas there are other possible aspects of Collatz sequences worth exploring.

15. Conclusion

In this study, I examined the behavior of Collatz sequences generated from all integers from 1 to 10,000 using Python. Stopping time, maximum value of each sequence and the ratio between the maximum value and the initial number were taken into account.

It appeared that the behavior of Collatz sequences differs greatly. The longest stopping time was recorded for the number 6171 which took 261 steps and the biggest maximum value for the number 9663 with 27,114,424 which is around 2,806 times bigger than the number itself.

The average stopping time was 84.97 steps and the median was 73 steps. Correlation coefficient between starting number and stopping time is 0.2054 showing weak linear relationship.

Thus, the experiment showed that the value of the starting number is not enough to predict how the sequence would behave. Though the study is not proving the Collatz Conjecture, it shows how computer science can be applied to an open problem in mathematics.

16. Future Scope

The project can be further expanded by experimenting with many more initial values, studying the possibility of continuation of the observed patterns in the range from 1 to 10,000 on a much greater scale. One more way of development for the program would be its optimization for the processing of a large amount of data.

It would also be useful to study various types of starting numbers, such as even and odd numbers, prime numbers and powers of two, as well as reasons for similarities in maximum values achieved and merging of sequences.

There can also be conducted the statistical analysis of the correlation between the value of the starting number, stopping time, maximum value and maximum to starting value ratio.

17. Acknowledgments

I would like to thank my teachers and family for their support and encouragement during this project.

18. References

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19. Appendix

Appendix A: Python Program Used to Generate the Dataset

```
import csv

with open("collatz_data.csv", "w", newline="") as file:

    writer = csv.writer(file)

    writer.writerow([
        "Starting Number",
        "Steps",
        "Maximum Value"
    ])

    for a in range(1, 10001):

        c = a
        d = 0
        Max = a

        while c != 1:

            if c % 2 == 0:
                c = c // 2
            else:
                c = 3 * c + 1

            d = d + 1

            if c > Max:
                Max = c

        writer.writerow([a, d, Max])

print("Dataset generated successfully!")
```

Appendix B: Data Analysis Program

```
import csv
import statistics
import math

data = []

with open("collatz_data.csv", "r") as file:

    reader = csv.DictReader(file)

    for row in reader:

        number = int(row["Starting Number"])
        steps = int(row["Steps"])
        maximum = int(row["Maximum Value"])

        ratio = maximum / number

        data.append({
            "number": number,
            "steps": steps,
            "maximum": maximum,
            "ratio": ratio
        })

steps = [x["steps"] for x in data]
maximums = [x["maximum"] for x in data]
ratios = [x["ratio"] for x in data]

print("Mean stopping time:", statistics.mean(steps))
print("Median stopping time:", statistics.median(steps))

print("Mean maximum value:", statistics.mean(maximums))
print("Median maximum value:", statistics.median(maximums))

print("Mean maximum/starting ratio:", statistics.mean(ratios))
print("Median maximum/starting ratio:", statistics.median(ratios))
```

Appendix C: Correlation Calculation

```
numbers = [x["number"] for x in data]
steps = [x["steps"] for x in data]

mean_x = sum(numbers) / len(numbers)
mean_y = sum(steps) / len(steps)

numerator = 0
denominator_x = 0
denominator_y = 0

for i in range(len(numbers)):

    x = numbers[i]
    y = steps[i]

    numerator += (x - mean_x) * (y - mean_y)
    denominator_x += (x - mean_x) ** 2
    denominator_y += (y - mean_y) ** 2

correlation = numerator / math.sqrt(
    denominator_x * denominator_y
)

print("Correlation:", correlation)
```

The program terminates for every tested starting value, but termination for every positive integer is exactly what the Collatz Conjecture asserts and has not been mathematically proved.

Appendix D: Important Computational Results

<u>Measurement</u>	<u>Result</u>
Starting values tested	10,000
Greatest total stopping time	261 steps
Starting number with greatest total stopping time	6171
Greatest maximum value	27,114,424
Starting number producing greatest maximum	9663
Highest maximum-to-starting-value ratio	2806.0048×
Mean total stopping time	84.97
Median total stopping time	73
Mean maximum value	58,996.48
Median maximum value	13,336
Mean maximum-to-starting-value ratio	11.79×
Median maximum-to-starting-value ratio	3.38×
Pearson correlation (r)	0.205405
Squared correlation (r^2)	0.0422